Girish Nalawade

girishnalawade020@gmail.com · (480) 406-1376 · Tempe, Arizona · LinkedIn · github.com/Girish-del

PROFESSIONAL SUMMARY

Results-driven Software Engineer and Network Engineer with 2+ years of experience designing and deploying distributed backend systems, cloud-native microservices, and AI-powered applications. Proven track record reducing latency by 30%, cutting deployment time by 40%, and improving system uptime to 99.9% in production environments. Skilled in Python, Java, Spring Boot, AWS, Kubernetes, and LLM integration — with hands-on experience in RAG pipelines, CI/CD automation, and Agile delivery.

WORK EXPERIENCE

Network Engineer — Arizona State University

May 2025 – Present | Tempe, AZ

- Engineered observability dashboards using Grafana and AWS CloudWatch, reducing mean time to detect (MTTD) incidents by 45% and lowering unplanned downtime across production storage clusters by 20%.
- Automated network configuration rollouts with Python and SNMP scripting, eliminating manual deployment steps and improving cluster uptime from 97% to 99.9% across distributed infrastructure.
- Hardened Cisco firewalls, VPN tunnels, and managed switches, cutting security-related downtime by 30% while sustaining full compliance with institutional network policies.

Associate Software Engineer — Western Union

August 2023 – August 2024 | Pune, MH

- Optimized backend transaction engine using Spring Boot, Redis, and asynchronous message queues, cutting payment processing latency by 30% (200ms → 140ms) across global payment networks handling millions of transactions daily.
- Co-built T-View, a real-time distributed monitoring system for 1M+ daily transactions, reducing fault recovery time by 40% through enhanced observability, health metrics, and alert pipelines.
- Architected containerized microservices using Kubernetes and Docker, cutting deployment time by 40% and improving cluster recovery speed under failure scenarios by 25%.
- Developed real-time event-driven services using WebSockets and Apache Kafka for push notification delivery, boosting customer engagement by 15% and reducing notification delivery latency by 22%.
- Automated CI/CD pipelines with Spinnaker, CloudBees, and AWS X-Ray tracing, ensuring secure, auditable, and compliant rollout standards across 10+ microservices.

Software Engineering Intern — Western Union

January 2023 – August 2023 | Pune, MH

- Led Spring Boot framework migration from v1.9 to v2.1 for 5+ backend services, modernizing dependency configurations and improving application security posture against known CVEs.
- Improved a serverless log archival system using AWS Lambda, S3, and CloudWatch, cutting storage costs by 35% and improving distributed traceability across 20+ microservices.
- Collaborated with QA and DevOps to integrate automated regression test suites into CI/CD pipelines, accelerating release delivery cycles by 20%.
- Executed load testing with JMeter under 500K concurrent requests, identifying and resolving 3 critical bottlenecks — improving fault tolerance by 33% and strengthening system resiliency.

Software Developer — Suvidha Foundation

October 2021 – May 2022 | Pune, MH

- Built resilient REST APIs using Spring Boot and Flask, achieving 99.95% uptime and enabling seamless integration between distributed microservices and enterprise client platforms.

EDUCATION

Arizona State University — Tempe, AZ

M.S. in Computer Science (Software Engineering) | Expected May 2026

Savitribai Phule Pune University — Pune, MH

B.E. in Computer Engineering | May 2023

PROJECTS

DevOps MCP Server – Model Context Protocol Integration for CI/CD Tooling

February 2025 – April 2025 | Python, FastAPI, MCP SDK, Docker, GitHub Actions, AWS

- Built a production-grade MCP (Model Context Protocol) server exposing 12 CI/CD tools — including GitHub Actions trigger, AWS ECS deployment status, and CloudWatch log streaming — enabling LLM agents to autonomously orchestrate deployments, cutting manual DevOps intervention by 60%.
- Implemented OAuth 2.0 authentication and tool-level access control for the MCP server, enabling secure multi-tenant usage across 3 teams with zero credential-sharing incidents.
- Reduced mean time to deploy (MTTD) by 45% by allowing AI agents to call deployment tools directly via structured MCP tool-use — eliminating context-switching between dashboards and CLI workflows.

MACE – Multi-Agent Coordination Engine

January 2025 – April 2025 | Python, FastAPI, LangGraph, Claude API, FAISS, React, Docker, GCP

- Architected a multi-agent orchestration system using LangGraph and Claude API that parses natural-language requests, routes subtasks to specialized agents (Support + Domain), and resolves resource conflicts via rule-based and LLM-powered arbitration — achieving 80%+ task completion rate vs. 50% for single-agent baseline.
- Built a FAISS-powered duplicate intent detection pipeline using sentence-transformers embeddings, achieving Precision@5 of 0.87 — reducing redundant agent calls by 35% compared to exact-string-match baseline.
- Designed a persistent shared memory layer (task registry, agent status log) enabling real-time agent coordination across concurrent sessions, with JWT-secured multi-user access and Docker Compose deployment on GCP Cloud Run.

TECHNICAL SKILLS

Languages:	Python, Java, Go, TypeScript, JavaScript, SQL, C/C++
Backend & Frameworks:	Spring Boot, Flask, FastAPI, REST APIs, gRPC, WebSockets, Microservices Architecture
Cloud & Infrastructure:	AWS (S3, ECS, Lambda, IAM, X-Ray, CloudWatch), GCP, Docker, Kubernetes, Terraform, OpenStack
Data & Messaging:	Apache Kafka, Spark, HDFS, Cassandra, PostgreSQL, MongoDB, Elasticsearch, Redis, pgvector
AI / ML:	LangChain, RAG Pipelines, LLM Integration (Gemini, OpenAI), sentence-transformers, FAISS, AST Parsing
Monitoring & Testing:	Grafana, CloudWatch, JMeter, pytest, Unit/Integration Testing, CI/CD (Spinnaker, CloudBees, GitHub Actions)
Collaboration:	Git, GitHub, Agile/Scrum, System Design, High Availability, Fault Tolerance, Scalability